

NANYANG JUNIOR COLLEGE

2008 JC2 PRELIMINARY EXAMINATION

MATHEMATICS

Higher 2

9740/02

15th September 2008

Paper 2

3 Hours

Additional Materials: Answer Paper

List of Formulae (MF 15)

READ THESE INSTRUCTIONS FIRST

Write your name, class and name of subject tutors in the space provided, as well as on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

NAME: _____

CLASS: _____ **SUBJECT TUTOR:** _____

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1		5		9	
2		6		10	
3		7		11	
4		8		TOTAL	

Section A: Pure Mathematics

- 1 Show that the differential equation $x^2 \frac{dy}{dx} - 2xy + 3 = 0$ may be reduced by means of the substitution $y = ux^2$ to $\frac{du}{dx} = -\frac{3}{x^4}$. [3]

Hence, or otherwise, find the general solution for y in terms of x . [3]

Sketch, for $x > 0$, two members of the family of solution curves corresponding to a positive value and a negative value of the arbitrary constant. [2]

- 2 The functions f and g are defined by

$$f : x \rightarrow \ln(x + 3), \quad x > -3$$

$$g : x \rightarrow -\frac{1}{4}x^2 + x, \quad -2 < x < 2$$

- (i) Show by means of a graphical argument that g is one-one, and find g^{-1} in similar form. [4]
 (ii) Show that fg exists, and state its rule and domain. [3]
 (iii) By finding the range of fg , or otherwise, determine if f^2g exists. [3]

- 3 A curve C is defined by the equation $2y \frac{dy}{dx} = y^2 - 1$ and $(0, 2)$ is a point on C .

- (i) Find the equation of the normal to the curve at the point $(0, 2)$. [2]

- (ii) Show that at $x = 0$, $\frac{d^3y}{dx^3} = \frac{57}{256}$.

Find the Maclaurin's series of y up to and including the term in x^3 . [6]

- (iii) Hence find the series expansion of e^y , up to and including the term in x^2 . [3]

- 4 A sequence of numbers, x_n , satisfy the relation

$$x_{n+1} = \frac{1 - x_n}{x_n - 4}, \text{ for } n \in \mathbb{A}^+.$$

- (i) If the sequence converges to a number L , show that L satisfies the equation $L^2 - 3L - 1 = 0$. [2]

- (ii) Obtain the exact value of the roots α and β of the equation $x^2 - 3x - 1 = 0$, where $\alpha < \beta$. [2]

- (iii) If $x_1 < 0$, find the value of L . [2]

- (iv) Show that $x_{n+1} - L = \frac{(L+1)(L-x_n)}{x_n-4}$. Deduce that $|x_{n+1} - L| < \frac{1}{5}|x_n - L|$ if $x_n < L$. [3]

- (v) Find a value N such that $|x_n - L| < 10^{-20}$ for all $n \geq N$ where $x_1 = -1$. [2]

Section B: Statistics

- 5 There are 10 members in a Clean and Green Committee made up of 5 married couples.
- (i) The committee members are to be seated at a round table for a meeting. There are 3 members who form the executive subcommittee. Find the number of ways the 10 members can be seated if
 - (a) each couple must sit together, [2]
 - (b) no 2 members of that subcommittee sit next to each other. [2]
 - (ii) The committee has to take a photo after the meeting. Find the number of ways they can be arranged in a row if all the men are together while all the women are together, and no husband must be next to his wife. [2]
- 6 Five girls Mary, June, April, Alice and Deborah, and five boys Matthew, Mark, Luke, John and Paul, stand in a queue in random order to enter the ballroom for Dinner and Dance 2008 – Splendour. Find the probability that
- (i) Matthew is first and Mary is sixth in the queue, [1]
 - (ii) either Alice is first or Mark is second (or both) in the queue, [2]
 - (iii) there is only one girl chosen if five people are chosen at random from this group, [2]
 - (iv) all five girls are next to each other given that at least four boys stand next to each other. [3]
- 7 A baker, on average, sells 4 out of 5 muffins a day. It is known that he prepares enough ingredients to sell exactly 200 muffins a day and that the cost price of the ingredients for 200 muffins is \$270. In any randomly chosen day, find the probability that
- (a) he can sell between 160 to 195 muffins (inclusive), [2]
 - (b) he can make a profit if he were to sell the muffins at \$1.80 each. [3]
- By using a suitable approximation, calculate the probability of the mean number of muffins sold per day to be at least 159 over a period of 60 days. [3]

- 8** A petrol station at Serangoon Gardens is opened 24 hours daily. On average, the number of cars arriving at the station is 7 in every 20 mins.

- (i) Find the probability that at most 5 cars will arrive at the petrol station during a 10- minute period; [2]
- (ii) Using a suitable approximation, find the probability that there are at least 500 cars arriving at the petrol station on a particular day. [3]

The petrol station will employ more staff if the probability that there are more than k 10-minute intervals in a day where more than 5 cars arrive at the petrol station is less than 0.05. Find the least value of k . [3]

- 9** (a) Nanyang Hotel employs 450 hotel staff consisting of 360 general workers, 30 administrative staff and 60 technicians. A random sample of 15 employees is to be chosen to represent the views of all the employees about the hotel. This sample is to be made so that there is as close a representation of all the staff as possible without bias to any particular individual or group. Explain how this can be done, justifying your working. [2]
- (b) A light bulbs manufacturing factory claims that the average life span of a particular type of light bulb is 1000 hours. The life span of the light bulb is known to follow a normal distribution. The consumer association believes that this is an overestimate and decides to test a random sample of 64 bulbs. The results are summarized as follows:

$$\sum x = 63\,910.4, \quad \sum x^2 = 63\,824\,061.$$

- (i) Calculate the unbiased estimates of the population mean and population variance of the life span of that particular type of light bulbs to four significant figures. [2]
- (ii) Determine, to one decimal place, the smallest value of the significance level of an appropriate test so that the sample provides sufficient evidence that the factory has indeed overestimated the life span of that particular type of light bulbs. [3]
- (iii) Find the range of values for the average life span the manufacturer has to claim so that it will not be rejected as an underestimate at 8% level of significance, leaving your answers to 3 decimal places. [3]

- 10** The weights of boys in a certain age group are normally distributed, with mean 52 kg and standard deviation σ kg. The weights of girls in the same age group are normally distributed, with mean μ kg and standard deviation 5 kg. On average, 1 in 25 randomly chosen boys weighs less than 45 kg; and 2 in 25 randomly chosen girls weigh more than 49 kg.

- (i) Find the value of μ and σ correct to 3 significant figures. [2]
- (ii) Find the probability that the weight of two randomly chosen boys is more than thrice the weight of a randomly chosen girl. [2]
- (iii) Find the probability that the mean weight of 10 girls chosen is less than 41 kg. [2]
- (iv) 50 boys are chosen randomly. Using a suitable approximation, find the probability that at least 46 boys are heavier than 46 kg. [3]

- 11** A student wishes to determine the relationship between the length of a pendulum, l , and the corresponding period, T . After conducting the experiment, he obtained the following set of data:

l / cm	150	135	120	105	90	75	60	45	30	15
T / s	2.45	2.31	2.22	2.07	1.91	1.74	1.56	1.35	1.10	0.779

- (i) Obtain a scatter plot of this set of data. [2]
- (ii) The student proposes the following two models.

$$A: T = a + b \ln(l)$$

$$B: T^2 = a + bl$$

- (a) Calculate the product moment correlation coefficients for both models, giving your answers to 4 decimal places. [2]
- (b) Determine which model is appropriate for this data. [1]
- (iii) Using the model determined in (ii) part (b), estimate the value of l when $T = 3$ s to 1 decimal places. Comment on the suitability of this method. [3]
- (iv) Find a value of l and its corresponding value of T such that the equation of the regression line for the chosen model will remain the same after the addition of this pair of values. [3]

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